Quotient reconstruction: native target preparation

isogeny-labs/crypto · 2026-10-26 · source caec72812

The next target-side computation is implemented. **The new target frame and 32-point challenge have not run.** Their inputs and acceptance gates are unchanged.

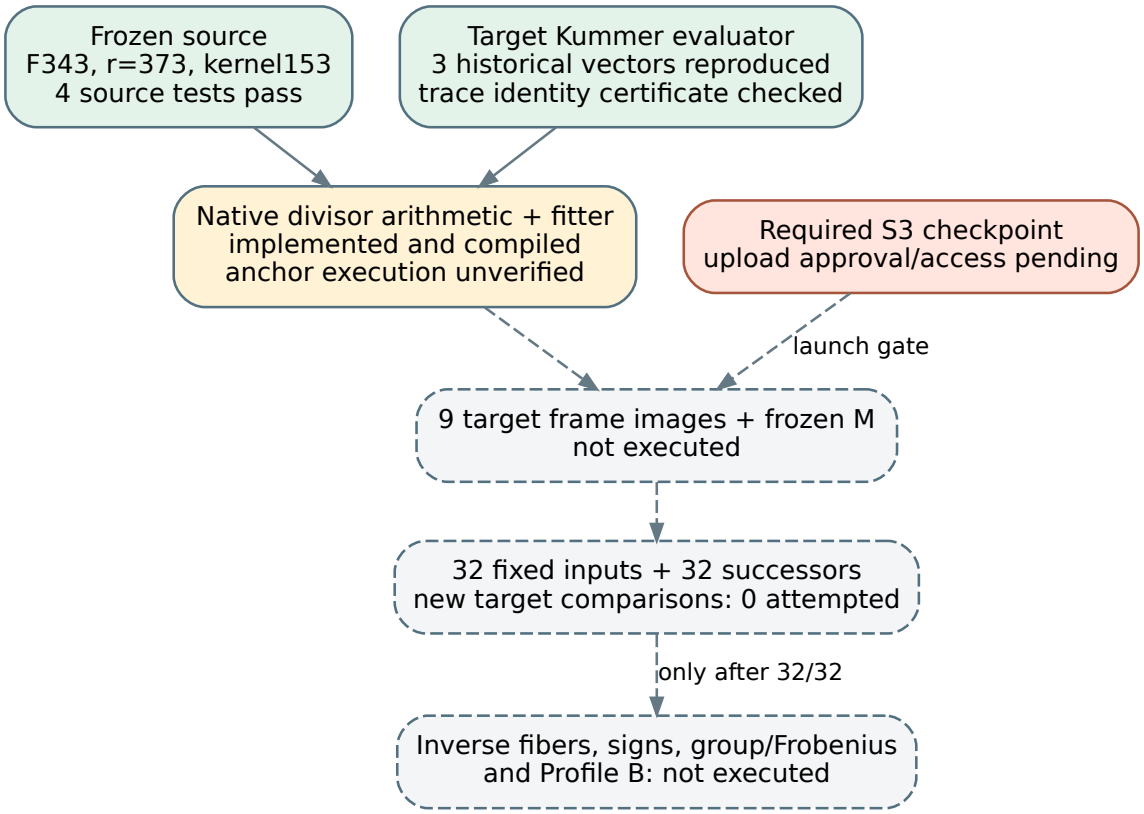

Figure 1: Green: completed fixed-component checks. Amber: implemented but unverified on the certified anchor. Dashed: unexecuted dependent gates.

Verified fixed components

Check	Observed result
Source package	4 tests pass, including full replay of the unchanged frozen source manifest.
Target package	5 component tests and 9 canonical certificate tests pass. Strict Clippy passes.
Historical targets	3 of 3 F49 Kummer vectors reproduced; direct six-permutation and residue-algebra evaluation agree.
Trace identity	IDC1h368ff5f1ba01b556: 32 derived and 16 extra field points accepted; exact integer expansion agrees on six monomials.

Implementation and scope

Khuri-Makdisi section spaces compute divisors over the explicit $F7^{18}$ field. An L-linear trace in a length-three residue algebra evaluates Tian's symmetric tensors without assuming their support splits in L. Full field coefficients are preserved. Calibration, source-only evaluation and the validation oracle have separate interfaces and fresh-only outputs.

The divisor engine's certified-anchor execution is unverified. No target matrix, inverse fiber, recovered divisor sign, global section identity, Profile B initialization or speedup is claimed.

Launch prerequisite

The official S3 checkpoint gate fails without configured upload access. Automatic approval review rejected reuse of the pasted key for the new checkpoint payload and destination; the action was not executed or retried. A review-only local snapshot was prepared and checkpoint approval/access requested. **No new target experiment launched.**

After access: freeze the native execution record, checkpoint it, evaluate the nine frame images, freeze/checkpoint M and require every one of the unchanged 32 primary and 32 successor comparisons before inversion. Preserve failures; no refit or replacement.

Sources: Tian, arXiv:2012.07173v1 section 3.2; Khuri-Makdisi, math/0105182 Lemmas 2.1–2.3; pinned historical reports in PROVENANCE.json. Detailed evidence, test output and limits: REPORT.md beside this PDF. No formal-prover or independent-runner review; no canonical cost-graph changes.